



CHAPTER 9 EXERCISES

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In this book, I have mentioned a lot of different exercises. It is imperative that you realize one thing: there are no essential exercises. For the most-part, my exercise recommendations are pulled from a menu of *movement patterns*, from which one can select any number of exercises. For example, the bench press, dumbbell overhead press, push-up, and dip are all upper body pressing exercises, and for most athletes are somewhat interchangeable.

Since this is a program design book and not an exercise technique book, our visitation to the details of the exercises will be brief. Exercises mentioned only once or used in examples may not appear in this list. Charlie Manganiello and I went quite deep into strength training in *Unstoppable Force*, which is a great starting point for learning some of these movements. Every single exercise listed in this book has a video associated with it online. Several of them are from us, several from others...caveat emptor.

Below are the primary exercises I have discussed above. Consider the following descriptions as starting points in your learning process of the movements. If you're not sure, hire a strength coach to help you. Most of these people really do know what they are doing.

AB WHEEL ROLL-OUT

The roll-out as performed correctly with the ab wheel is one of the most effective abdominal strength exercises you'll do. Position yourself with knees on a pad, pre-contract the abs so that you move into the "cat" position, then with the wheel on the floor in

front of the knees, roll out the the end of your comfortable range of motion. A mistake we see with many athletes is to go too far into the movement, and form collapses on the return to the start. Be sure to “press” the wheel into the ground on the return, and avoid leading the backward motion with the hips. The movement should not feature bending at the hips at any point. If you can do more than a few reps, intensify the effort by slowing down.



AB Wheel Start



AB Wheel Bottom

ARM BAR

The kettlebell arm bar is a shoulder mobility and stability drill. Lying on the floor, press a light kettlebell up to a straight arm position, keeping your grip tight and wrist straight. Place the opposite hand overhead, in line with your body. Rotate your body to the straight-arm (unloaded) side, maintaining a stable position with the kettlebell above you. Rotate your hips until your top-side knee comes down to the floor in front of you, then continue the rotation until you feel a significant stretch across the front of the loaded shoulder.

Both arms will be stretched in this position: the loaded arm’s pectoral muscle and the unloaded arm’s lat will both be elongated. Increase the lat stretch by crawling the fingers “backward” across the floor. Increase the loaded arm stretch by rotating the hips toward the floor.

You don’t need to use a lot of weight here - a heavier bell won’t speed your progress. Start with around 30 seconds per side, working up to a minute as your shoulder stability increases.



Arm Bar Start



Arm Bar Middle



Arm Bar Top

BALL SLAM

The ball slam is best performed with a sand-filled rubber “jam ball,” but can also be done with a standard medicine ball. Jam balls range in weight from 5-80 pounds, and don’t rebound when they hit the floor. The exercise is performed by bringing the ball to full extension overhead with both hands, then throwing it with maximum force into the ground. We like to use 10-20 pound balls, as a heavier weight slows down most athletes and tires the shoulders unnecessarily. The focus should be on maximum speed of movement more than high reps or high loads.



Ball Slam Start



Ball Slam Bottom

BENCH PRESS

The bench press is a well-known upper body press. We normally perform this exercise with a barbell or pair of dumbbells. The weight should be gripped at approximately shoulder-width apart,

but can be adjusted for greatest comfort. Gripping the bar tightly, lower the weight to the chest then return it to the top position. The bench press is a heavily-studied movement and you will see many variants and coaching cues. For our purposes, we simply suggest keeping the elbows held as close as possible to the ribs as you descend and always performing the full range of motion. Keep the feet planted firmly on the floor at all times.



Bench Press Bottom



Bench Press Top

BOX JUMP

The box jump is a simple exercise and is a staple of conditioning programs. It is also a great tool for power development. You should keep in mind that the height of the box is less important than how high you jump. To perform a proper power box jump, the athlete starts in a half-squat position (your natural launch depth - don't overthink it), then explodes upward onto a stable box or bench. The landing should be executed in the same position as the take-off, meaning that landing in a deep squat is a result of too little power. If you can't land in the same position, reduce the height of the box. Fatigue is not a part of this exercise, and each rep should be about quality and power more than sweat and soreness. Step down, as opposed to jumping down, between each rep. If you can do all the prescribed reps on the highest box available easily, consider doing single-leg jumps.



Box Jump Start



Box Jump Middle

CAMPUS LADDER

The Campus Ladder is a staple of finger power work. Simply climb up the board hand over hand. Clearly, if this is a challenge for you, you need to consider regressing the movement for the time-being. The first progression is to match each rung, followed by laddering left then right one hand per rung. As your strength increases you can skip rungs as your ability allows. In power phases, the general idea is to aim for more explosive movements and not worry much about reducing rung size. In strength phases, reducing rung size is a normal part of progressing. Keep in mind that this is a very intense exercise, and you should always try to keep your ego out of it.



Campus Ladder

DEADLIFT

The deadlift is the ultimate strength exercise. In this program, I recommend the standard deadlift, done with a barbell and a double-overhand grip. As climbers, we want to do this exercise without the aid of wrist straps or belts, as two of the major strengths we are trying to develop are the grip and the tension created by the midsection.

The stance should be shoulder width apart or slightly narrower with your toes slightly kicked outward or neutral. The hips should be above your knees and your shoulders above your hips. The shins should be vertical and the barbell should be touching your shins. The grip is outside your stance, somewhere around shoulder width apart. The head, neck, and spine are neutral.

Before you lift, take the “slack” out of the bar. This means tense your whole body like you’re going to lift the bar, but don’t. Have a proud chest, squeeze the barbell, flex your lats, and fire your glutes. As you pull the barbell up your shins, the hips and shoulders rise simultaneously while keeping a neutral spine. At the top, completely lock out the deadlift by squeezing your glutes and your quads. Reverse the movement and once the barbell gets past your knees you can drop it to the floor.

If you don’t have experience with this lift, it is well worth opening your wallet to get some instruction.



Deadlift Bottom



Deadlift Top

DRAGON FLAG

The Dragon Flag is a top-end midsection exercise. This is best done on a flat bench, but can be done on the floor with the hands grasping a post or other fixed object. The idea is to raise the entire torso and legs using the shoulders as a fulcrum, and then lower back to the start position under control. Control - it's the whole game in hard midsection work. Don't cheat it, and don't hurry it.

Most of us can't do a proper dragon flag until months into training it...so what to do? We progress it through a pike position and slowly move toward proper movement. To do this, start with your back on the bench and grasping the bench with the hands on either side of your head. Bend at the hips, holding your legs perpendicular to the bench, feet toward the ceiling. From here, raise the hips from the bench, and slowly straighten the body until at full extension, with just the shoulders and head (and hands) in contact with the bench.

From here, you can either reverse the pike or you can work toward lowering your body into the straight flag position. As far as progression, this pike-to-lowering strategy works very well to help you gain the proper form. Once you can do a solid five reps, slow the movement down by half and work up to five again.



Dragon Flag Start



Dragon Flag Middle



Dragon Flag Bottom

DUMBBELL ROW

This exercise is the workhorse horizontal pull exercise. It can be a very basic, lightweight rehab movement, and can be a heavy power movement, as well. The setup is to use a flat bench, with one hand and the same-side knee placed on the bench. With the opposite hand, you hold a dumbbell straight down at arm's length. Under control and without rolling the shoulder, pull the dumbbell to the side of the ribs, then return it to the start position. Start with your non-dominant side, then follow with the stronger side.



Dumbbell Row Start



Dumbbell Row Top

FROG STRETCH

The frog drill is a hip-opener that will help not only with improving your squat form, but with your turn-out on the rock. The frog is performed prone, with the knees turned out to the sides, and placed on a mat. Most knees can't handle this exercise on a hard floor.

1. Turn the feet out to the sides, and place the elbows on the ground directly below the shoulders. Most of your weight should be on your arms.
2. Rock the hips back toward the heels, then move them slowly back forward. Once at the top position, try to move the knees out to the sides slightly, then repeat the motion.

3. Maintain the movement slowly for 45-60 seconds, trying each time to relax into the bottom position.



Frog Stretch Start



Frog Stretch Top

FRONT SQUAT

The squat is a fundamental human movement and is the best exercise for developing leg strength. Beyond the legs, the front squat develops the ability to pressurize the core and build mobility in the hips. For this exercise we prefer the kettlebell or barbell... dumbbells don't seem to sit right. The first rule of squatting is to master the range of motion. Squatting down “between your heels” and maintaining a neutral spine, you'll want to dip to where your hip joint is below the knee. Your heels should remain planted to the floor at all times, and your knees should track the toes. On the return ascent, the hips and shoulders should ascend at the same rate.



Front Squat Start



Front Squat Bottom

HANGBOARD - FULL CRIMP

The full crimp position is the most dangerous position we train, but it's the one we revert to when the chips are down on a route or boulder. We train these conservatively, and progress slowly. This grip features a closed-handed position where the second knuckle (PIP) is held above the level of the hold.



Full Crimp

HANGBOARD - HALF CRIMP

The half crimp is the most common hold position, and is the safest grip for most climbers. In this position, the distal phalanges are held more-or-less flat, with the PIP joint (second knuckle) kept at the same level as the fingertips.



Half Crimp

HANGBOARD - OPEN HAND

In the open hand position, the second joint (PIP) is held at a level below the fingertips. This is the position recommended for holding large holds and pockets in most cases, and is quite low-stress on the finger joints.



Open Hand

HANGBOARD - PINCH

The pinch grip is a fundamental hold position for many climbs, yet is difficult to train in isolation. Today's hangboards fall short of effective in training this grip position, and most of us don't have access to a good system wall on which to train pinches. For the hangboard pinch position, we have found that pinch blocks work best. These blocks, somewhere between 1.5" and 3" thick can be loaded with weight and deadlifted to train the pinch position.



Pinch Block



Loaded Pinch Block

HANG, BENT ARM

This single bent-arm hang is done holding the half-crimp position, with the elbow bent to about 90 degrees. Using a thin cord or loop of webbing, take weight off with the other hand until you can hold the position for the planned duration. Over time, you can progress to holding bodyweight for the duration. Once you can hold bodyweight, it's time to advance to a smaller edge, and go back to assisted hangs. Start each set with the “strong” side, followed by the “stronger” side.



Bent Arm

HANG, LOCK OFF

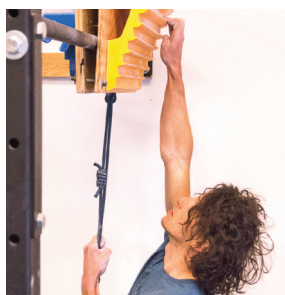
The lock off one-arm hang is performed with as closed an elbow joint as you can manage. Fingers are maintained in the half-crimp position. Using a thin cord or loop of webbing, take weight off with the other hand until you can hold the position for the planned duration. Over time, you can progress to holding bodyweight for the duration. Once you can hold bodyweight, it's time to advance to a smaller edge and go back to assisted hangs. Start each set with the “strong” side, followed by the “stronger” side.



Lock Off

HANG, STRAIGHT ARM

The single-arm straight-arm hang is performed with the fingers held in a half-crimp position and the elbow locked straight. The most potentially dangerous single-arm position, it is critical that you keep your shoulder engaged during this exercise. Think about drawing your shoulderblade “down” and keeping your elbow facing forward. As with the other single arm exercises, you can use a piece of webbing held in the other hand to take some weight off as you adapt to this load. Start each set with the “strong” side, followed by the “stronger” side. Once you can hold bodyweight, it’s time to advance to a smaller edge and go back to assisted hangs.



Lock Off

HANGING STRAIGHT LEG RAISE

The hanging straight leg raise, or HSLR, is done by holding on to a pull-up bar that is high enough for your feet to swing freely below. With packed shoulders (elbows “forward”) move from a straight position to bring your legs up in front of you. Keep the knees locked straight and control the movement at all times. Try to bring the legs above horizontal, then return to the bottom with strict control. Each rep should take 3-5 seconds, with no swinging at the bottom.



HSLR Start



HSLR Top

INVERTED ROW

The inverted row should be a staple of any upper body program. It is an excellent horizontal pull movement that has the added benefit of integrating core stability into the move. You can use any of several implements, but my favorite is a suspension trainer or set of rings. Begin with straight arms in a rigid position, effectively the opposite of the way you’d set up for the push-up. From the bottom, pull to a locked position with the hands at the sides of the ribs. If you can’t do the full range of motion, lower your foot position to where the move is smooth and complete. Variations include the single arm row, inverted rows on a fixed bar, and a single arm row with a rotation at the shoulder.



Inverted Row Start



Inverted Row Top

KETTLEBELL SNATCH

The kettlebell snatch is an excellent ballistic exercise. Starting in the standard “hike” position, you’ll swing a single bell backward as normal, but on the upswing you’ll power the bell up to a locked out position overhead. Be careful not to overswing the kettlebell. As the weight passes your torso, pull the arm back to “tame the arc” then push the hand straight up. This causes the bell to rotate easily around the wrist rather than flipping over the top.

This is a high-skill movement and should be practiced extensively before implementing it into your training. That being said, I believe that this is such a great exercise (once you can perform it well) that you really are missing out if you don’t take the time to learn it. In particular, the rhythm of the ballistic-to-static in this exercise is similar to the power output and timing of hard bouldering, which is difficult to simulate in the weight room.



KB Snatch Start



KB Snatch Middle



KB Snatch Top

KETTLEBELL SWING

We use this exercise with almost all of our athletes, almost all the time. The kettlebell swing is a great exercise for lower body power development and integration of movement. Start with the ‘bell on the ground a foot or more in front of you, then reach down and “hike” the bell back to start the reps. When swinging, only squat enough to get the bell between the legs (handle passing above the knees) and snap the hips forward to raise the kettlebell to the height of the sternum. Swinging higher in the so-called “American swing” actually diminishes the power component of the exercise, making it an easy knee-dominant movement instead of a hip dominant move. Many coaches caution against swinging overhead due to the dynamic lumbar flexion that most athletes exhibit at the top of the move.



KB Swing Start



KB Swing Middle



KB Swing Top

LEVER

The front lever is the top-end of core training. This movement requires good shoulder strength and stability, as well as a very strong midsection. Grasp a pull-up bar, then rotate your body to a horizontal position using your shoulder as a pivot point. Return to vertical to complete the rep.

It is critical that you exercise full control throughout the movement. If you can’t complete the level with both legs

held straight, try the move with one knee tucked to the chest, or even both tucked to the chest in a move that resembles “knees-to-elbows.”



Front Lever Start



Front Lever Top

ONE ARM OVERHEAD PRESS

Although push-ups are climbers’ go-to pressing exercise, the strict overhead press is what I consider the best bang for your buck. Pressing and holding weight overhead not only improves your shoulder stability and strength, but it is an excellent developer of core stability and strength. For this exercise, kettlebells are the superior tool. From the floor, you’ll clean the kettlebell into rack position, then press the weight overhead in a smooth movement with no dip in the knees. Repeat the press from rack (it is not necessary to re-clean the weights each rep) for the prescribed reps.

Climbers tend to have terrible overhead mobility, so an overhead press usually looks like a Nazi salute or an incline bench press with a huge arch in the low back. As much an exercise for mobility as for strength, there are some definite rules you should follow.

1. Keep your quads, glutes, and midsection tight or “flexed”. This keeps your back safe and helps you practice total body integration.
2. Keep your wrists neutral and “crush” the handle of the weight.

3. Lock out the elbows at the top of each rep and pause briefly in the top position.
4. Pause at the bottom for one or more seconds between each rep.



One Arm Press Start



One Arm Press Top

PLANK/PULL

The king of the core dynamic stability exercises, this one is done with a low cable or low-anchored elastic band. From an elbow plank position, grasp the band or cable handle, and pull it down along the body, much like the movement of a 1-arm pull-down. Our normal prescription is to switch hands every 5-8 reps (without leaving plank position) for a given duration, usually starting about a minute. As you'd guess, slow, controlled movements are key.



Plank/Pull Start



Plank/Pull Finish

PULL-UP

The pull-up is one of the classic upper body strength exercises. It is a landmark strength movement. Although highly bastardized into lesser movements, the strict pull-up is a great exercise. Start with the hands shoulder-width apart on an overhead bar, with elbows locked straight and shoulders in the “packed” position. In a smooth, SLOW movement, pull until your entire head is above the bar or until the bar touches the collarbones. The top of the range is the hardest, and thus should be the focus of your practice as you work toward your first full pull-up. Reverse the movement exactly: slow, controlled, and ending in the straight elbow, packed shoulder position. Regressions include (in order): 3 second deadhang at bottom, 3 second lock off at top, 1/4 chin up from top, 1/2 chin up from top, 3/4 chin up from top, and full chin. Progressions include tactical pull-up, power pull-up, L-Pull-Up, weighted pull-up, and offset variations.



Pull-Up Start



Pull-Up Top

RACK SQUAT

The rack squat is a front squat done with kettlebells. As simple as this variation may seem, changing to kettlebells represents a substantial improvement in the movement for most athletes. Positioning the kettlebells in “the rack” forces a solid core and allows you to move quickly between exercises: cleaning the

kettlebells and starting to squat is much faster and easier than setting up a squat bar.

Start with a pair of kettlebells placed side by side on the ground in front of you. Grab the handles and clean the bells to your shoulders (if you don't know the clean, now is the time to learn). In rack, you should be standing tall, bells sitting in the crooks of your elbows, with the elbows held tight against the ribs. Forearms should be vertical, and wrists straight.

The movement involves sitting “between the heels” and maintaining that good, upright position. This is an easier variant of the squat than most, and you should find full range of motion is easier than in the barbell variants. Look for a clean, smooth movement - if you have hitches in any part of the range of motion, back off a little on the weight. At the end of the set, reverse the clean (rather than dropping the bells in some weird side to side curl movement) and set the bells down.



Rack Squat Start



Rack Squat End

TRAP BAR DEADLIFT

The Trap Bar is a barbell with a diamond or hexagonal space in the middle, in which the lifter will stand . It allows the deadlift to be performed with less forward lean, and is often used as an instructional tool for new lifters. We also like trap bar deadlifting as a variation to the hard pulling of the traditional deadlift.

THE SET-UP:

Step inside the trap bar. With feet slightly narrower than shoulder width apart, hinge at the hips until you can reach the handles. The arm position on the trap bar is wide, so you'll need to drop your body lower than in a normal deadlift. Drive the hips back and try to maintain as vertical a shin as possible. Keep the neck neutral and eyes on the floor in front of you.

THE PULL:

Tighten the glutes, and with a “proud” chest pull the bar off the floor slowly. Jerking the bar tends to cause your back to lose stiffness. Although you'll be able to “squat” more in this variation, keep in mind that the end goal is to deadlift more effectively, so keep focusing on the hinge. Continue the pull until you are upright and the knees are locked straight. To reverse the move, inhale, set the abs, and drive it carefully back to the ground.



Trap Bar DL Start



Trap Bar DL Top

FINAL THOUGHTS



PHOTO: NATE LILES